Java Lab Assignments

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Problem Statement 1:

Matrix Arrangement:

Write a function to return an array where the midst position contains the smallest value followed by the next smallest value on the right of the midst position, followed by the next smallest value to the left of the midst position and the rest of numbers continue in this format

Function signature:

```
public static int [ ] arrangeElements(int[ ] [] inputArray) {
// write the code here

}
```

#	Sample Input	Sample Output
UTC_01_01	4 2 13 3 8 5 9 6 17	{17,9,6,4,2,3,5,8,13}
UTC_01_02	4 1 3 3 8 5 9 16 17	{17,8,4,3,1,3,4,5,16}

Steps:

1) Form eclipse IDE import -> General - >Existing projects into workspace and select root directory where you have extracted the provided assignments zip file
2) Select "Java_Day1_assignments"
3) Complete the static function [arrangeElements ( )] present in "ArrayTasks.java" and test them with the Junit tests provided in "ArrayTasksTest.java"

Problem Statement **2**

Sum of consecutive elements in a subsequence

Complete the method to find the minimal length of the subsequence of consecutive elements of the sequence, sum of which is greater or equal to the specified number.

Example:

5	1	3	5	10	7	4	9	2	8

If the specified number is 15 in the sequence

5 + 1 + 3 + 5 + 10 > 15 (length is 5)

1+3+5+10 >= 15 (length is 4)

3+5+10 >= 15 (length is 3)

5 + 10 >= 15 (length is 2)

10 + 7 >=15(length is 2)

There is no element in the array whose value is >= 15 and hence the minimal length is 2

```
public static int findMinimalLength(int[] array, int value){
    //code here
}
```

#	Sample Input	Sample Output
UTC_02_01	array = {5,1,3,5,10,7,4,9,2,8} value = 15	2
UTC_02_02	array = {1,2,3,4,5} value = 11	3

Steps:

1) Form eclipse IDE import -> General - >Existing projects into workspace and select root directory where you have extracted the provided assignments zip file
2) Select "Java_Day1_assignments"
3) Complete the static function [findMinimalLength ( )] present in "ArrayTasks.java" and test them with the Junit tests provided in "ArrayTasksTest.java"

Note: Add few more tests of your own to test the written code.

Problem Statement 3:

Guessing Game

Write a Java program to play the word guessing game.

The computer will pick a question and you must guess the answer. If you guess the word with less than MAXGUESS_COUNT mistakes, then you win.

The questions and answers should be stored in static array of String.

Use java.util.Scanner class to read input from keyboard.

The following suggestions might be helpful:

1. Program should choose a random question from a list of available questions.

   Sample code to generate Random Number:

```
public static int randInt(int min, int max) {
      java.util.Random rand = new java.util.Random();
      // nextInt is normally exclusive of the top value,
      // so add 1 to make it inclusive
      int randomNum = rand.nextInt((max - min) + 1) + min;
      return randomNum;
}
```

2. Program should hint how many characters are there in the answer by displaying "*" for every character. For example if the answer to the question is "ice cream", it should display "*** *****".
3. Each time the user makes a guess, ask the user if he/she would like to play another game.
4. The user is provided five chances to guess the correct answer; if he/she fails the program will be terminated.

Sample input/output:
Question: Man's best friend
The word has 3 letter(s).
The word is: * * *

What is your guess? You have 5 guesses remaining.
Enter: cat
Sorry, the word is not "cat".

The word is: * * *
What is your guess? You have 4 guesses remaining.
Enter: bat
Sorry, the word is not "bat".

The word is: * * *
What is your guess? You have 3 guesses remaining.
Enter: dog
Yes, the word is "dog"!
You made only 2 mistakes. Good!

Problem Statement 4:

Time

Complete the methods provided in Time.java to sum two time objects. Test the written code by executing TimeTest.java Junit tests

#	Sample Input	Sample Output
UTC_04_01_01	Time t1 [4 : 30] Time t2 [3 : 45]	Time [8 : 15]
UTC_04_01_02	Time t1 [2 : 30] Time t2 [0 : 15]	Time [2 : 45]
UTC_04_02_01	Time t1 [4 : 30] Time t2 [3 : 45]	Time [8 : 15]
UTC_04_02_02	Time t1 [2 : 30] Time t2 [0 : 15]	Time [2 : 45]

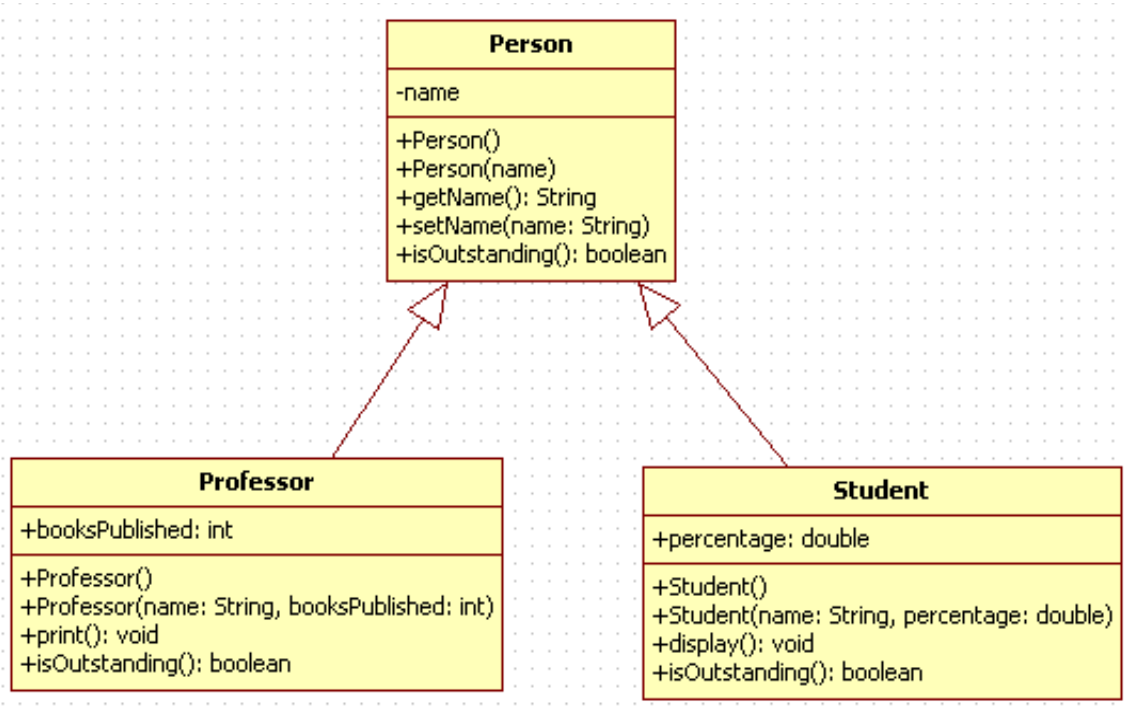
Steps:

1) Form eclipse IDE import -> General - >Existing projects into workspace and select root directory where you have extracted the provided assignments zip file
2) Select "Java_Day1_assignments"
3) Complete the static function [static Time addTime(Time first, Time second)] present in "Time.java" and non-static method[public Time addTime(Time other)]. Test them with the Junit tests provided in "TimeTest.java".
4) Provide additional methods if required

Problem Statement 5:

Outstanding persons

Outstanding: Distinguished from others in excellence.
We judge a person if he is outstanding or not by the achievement in his/her profession



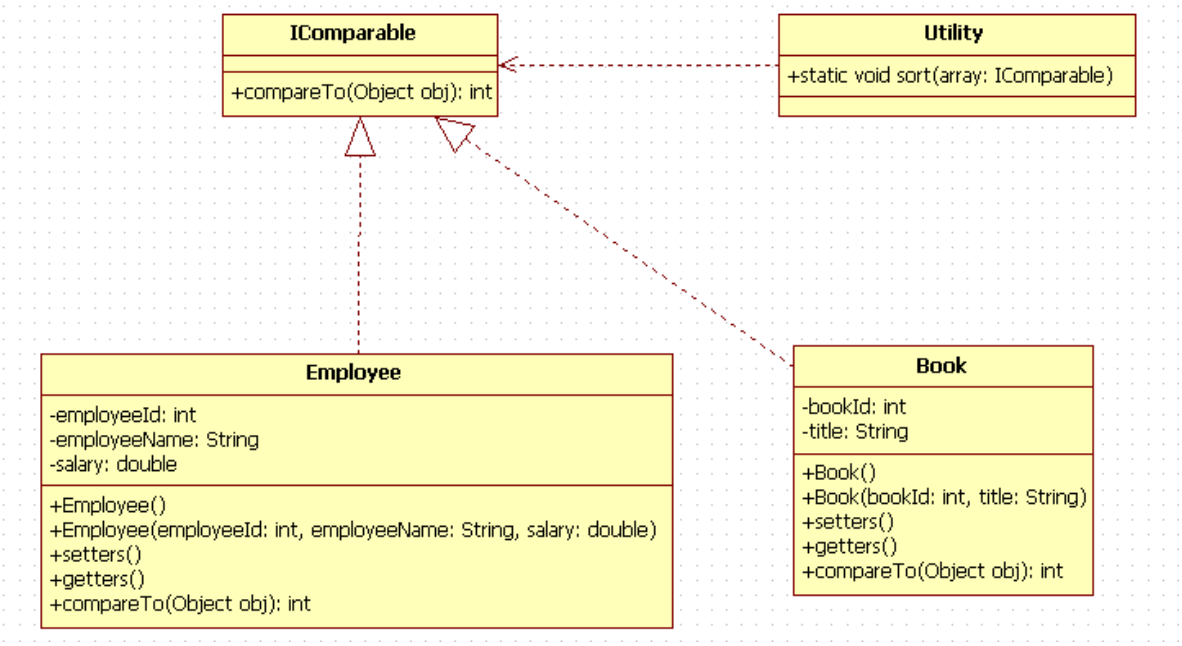
Write an application to perform the following operations:

1) Complete the entity classes shown in Class diagram
2) Implement isOutstanding( ) based on the following rules:
    a. Professor is outstanding if he has published more than 4 books
    b. Student is outstanding if his percentage is greater than or equal to 85.
3) The print ( ) method of Professor displays the name and books published by professor.
4) The display( ) method of Student displays the name and percentage of student
5) Complete the main ( ) method of PersonExample.java to implement the following:
    a. Store 5 persons in a single collection of array type (Person[ ]) , this can be a combination of students and professors
    b. Using Run-Time Type Identification print Professor and Student details.
    c. Identify all outstanding persons
6) Write Junit tests to test your code

Problem Statement 6:

SORTING DIFFERENT ENTITES

Sorting functionality using Open/Closed Principle: The functionality is programmed to accept an interface and is closed for modifications. If the functionality has to be used on any entity at a minimum it should implement that interface.



Write a program to perform the following operations:

1) Write the entity classes as shown in Class diagram
2) Complete the sort( ) method of com.visa.prj.utility.Utility to sort entities
3) Book should be sorted based on title and Employee should be sorted based on salary
4) Write main( ) method to test

Problem Statement 7:

Product collection

any online shopping portal, the products will be displayed in order. The user interface allows the products to display sorted by brand or by price; similarly in employee portal employees can sorted by designation, last name or first name, etc.

Create a Product entity class to hold the following data:

Product ID	Brand Name	Description	Price
200	Dell	15 inch Monitor	3400.44
120	Dell	Laptop	45000.00
150	Microsoft	Windows 7	7000.50
100	Logitech	Optical Mouse	540.00

Write a program to perform the following operations

1. Stores the data shown above in a List implementation.
2. An operation to display all the products in a sorted order by default based on their product id.
3. Option for sorting based on Brand name or price based on input provided at runtime.
   a. If two products contain same brand name, description should be considered.
   b. Similarly if products have same price, product id's should be considered while displaying them in order

Sample Output if Sorted by Brand Name:

Product ID	Brand Name	Description	Price
200	Dell	15 inch Monitor	3400.44
120	Dell	Laptop	45000.00
100	Logitech	Optical Mouse	540.00
150	Microsoft	Windows 7	7000.50

Problem Statement 8:

Best Selling Fruits

"BestFruits" is a popular fruits shop. As part of their business, they want to maintain a report of best selling fruits for their business improvements. So,every time a consumer purchases a fruit, the best selling fruits report should be updated with the count. The report also should maintain the fruit selection trend in the descending order of the sales, ie., the most selling fruit should be at the top of the report and the least selling fruit at the bottom.

Write an application to accept fruit selected by the customer and update its count and re-arrange the order based on the count. If the fruit is not present in the collection add it to the end of the collection with its count as 1.

Example:
If new fruit "Pear" is added fruit details sorted in descending order of count as shown below should be displayed:

Mango	121
Apple	60
Orange	35
Pear	1

Write JUnit test methods to test your code.

Problem Statement 9:

Regional Transport Offices of Bangalore and Vehicle Registrations

Description:

Vehicles registration's first two parts "XX-XX" contains the state information and the RTO zonal area code were the vehicle was registered.

For example if vehicle registration number is "KA-50-EF-1234", then "KA" refers to "Karnataka" and "50" refers to Zonal office area "Yelahanka" and similarly if the vehicle registration is "KA-05-45", then "KA" refers to "Karnataka" and "05" refers to Zonal office area "Jayanagar".

Given below is the complete set of RTO Code and corresponding area names of RTO's in Bangalore:

RTO Code	Area
KA-01	Koramangala
KA-02	Rajajinagar
KA-03	Indiranagar
KA-04	Yeshwanthpur
KA-05	Jayanagar
Ka-50	Yelahanka

RTO Code	Area
KA-51	Electronics City
KA-52	Nelamangala
KA-53	K.R.Puram
KA-54	Nagamangala
KA-55	Mysore East
KA-56	Basavakalyan
KA-57	Shantinagar

Complete a method which accepts a collection of registration numbers and returns a collection of registration numbers which are sorted on RTO-Zonal area where the vehicles were registered. Note: If vehicles are from same area then they should be sorted based on registration sequence number [Part of number without RTO Code]

UTC	Input	Expected Output	Description
UTC_14_01	KA-55-AB-4555, KA-01-EF-4444, KA-04-AB-9000, KA-56-200, KA-50-T-3111, KA-02-AG-9243	KA-56-200, KA-01-EF-4444, KA-55-AB-4555, KA-02-AG-9243, KA-50-T-3111, KA-04-AB-9000	Vehicles are sorted based on area namely: **Basavakalyan, Koramangala, Mysore East, Rajajinagar, Yelahanka, Yeshwanthpur**
UTC_14_02	KA-57-DE-111, KA-51-A-9, KA-04-500, KA-02-L-41	KA-51-A-9, KA-02-L-41, KA-57-DE-111, KA-04-500	Vehicles are sorted based on area namely: **Electronics City, Rajajinagar, Shantinagar, Yeshwantpur**
UTC_14_03	KA-57-DE-111, KA-51-A-9, KA-04-500, KA-02-L-41, KA-57-AB-9011, KA-04-A-100	KA-51-A-9, KA-02-L-41, KA-57-AB-9011, KA-57-DE-111, KA-04-500, KA-04-A-100	Since Both the vehicles "KA-57-AB-9011" and "KA-57-DE-111" are from same area they are sorted based on sequence. Similarly for "KA-04-500" and "KA-04-A-100"

Note: Use Appropriate Java Collection API to solve this problem statement.

Problem Statement 10:

Word Count

Given a sentence find the number of characters, words and lines in the sentence. Complete the provided method that counts the no. of Characters, Words and Lines in a file and returns the count in a Map.

Input: A sentence

Output: A Map returning the Character count, Word count and Line Count.

Test Case	**Input**	**Output**
UTC_15_01	All our dreams can come true, if we have the courage to pursue them	Character count :53 Word count :14 Line count :1
UTC_15_02	Great works are performed, not by strength, but by perseverance	Character count :52 Word count :10 Line count :2

Problem Statement 10:

Movie and Actors information.

Given data as listed below:

Movie and actor details			
Title	Studio	YEAR	Actors
Skyfall	MGM	2012	Daniel Craig, Javier Bardem, Ralph Fiennes
Schindler's List	Universal Pictures	1993	Liam Neeson, Ben Kingsley
Staying Alive	Paramount Pictures	1983	John Travolta
The Dark Knight Rises	Warner Bros	2012	Christian Bale, Michael Caine, Morgan Freeman
The Taking of Pelham 123	Columbia Pictures	2009	Denzel Washington, John Travolta
Octopussy	United Artists	1983	Roger Moore, Maud Adams
Jurassic Park	Universal Pictures	1993	Sam Neill, Laura Dern, Jeff Goldblum

Write the "MovieServiceImpl.java" implementation class of MovieService.java interface with method definitions declared in MovieService.java interface. This class should use appropriate java collection classes for storing and fetching movie/actor details as listed below.

The application has to do the following:

a) Store the above details using appropriate data structures provided by java collection framework.
b) Display movies details and actor names **sorted based on release year**, if there are many movies released in a year they should be sorted alphabetically as shown below:

Year: 1983
Movie: Octopussy
Studio: United Artists
Actors:
Roger Moore
Maud Adams

Movie: Staying Alive
Studio: Paramount Pictures
Actors:
John Travolta

Year: 1993
Movie: Jurassic Park
Studio: Universal Pictures
Actors:
Sam Neill
Laura Dern
Jeff Goldblum

Movie: Schindler's List
Studio: Universal Pictures
Actors:
Liam Neeson
Ben Kingsley

Year: 2009
Movie: The Taking of Pelham 123
Studio: Columbia Pictures
Actors:
Denzel Washington
John Travolta

etc…

c) Display Movies acted by a given actor. Example is shown below:

i) **Input: John Travolta**
Movies:
Staying Alive
The Taking of Pelham 123

ii) **Input: Ben Kingsley**
Movies:
Schindler's List

Problem Statement 11:

Cryptography

Cryptography: scramble the message.
Cryptography has two disciplines:
– Cryptography: developing codes/ciphers
– Cryptanalysis: breaking codes/ciphers

Symmetric Methods Columnar Transposition Cipher
To encrypt: Write the plaintext in columns of depth k (= key), padding (with *) the end as necessary, read off in rows
Note: Key can be between 1 and (message length -1)

Example if key = 4.
Plaintext: Transposition ciphers are easy!

T	s	i	N	p	s	e	s
r	p	t		h			y
a	o	i	c	e	a	e	!
n	s	o	i	r	r	a	*

Cipher text: Tsinpsesrpt h yaoiceae!nsoirra*

Example if key = 3.
Plaintext: Password Welcome123

P	s	r	W	c	e	3
a	w	d	e	o	1	*
s	o		l	m	2	*

Cipher text: PsrWce3awdeo1*so lm2*

Example if key = 2.
Plaintext: Caeser cipher

C	e	e		i	h	r
a	s	r	c	p	e	*

Cipher text: Cee ihrasrcpe*

Test Case	Input file	Output file
UTC01_01	Transposition ciphers are easy!	Tsinpsesrpt h yaoiceae!nsoirra*
UTC01_02	Password Welcome123	PsrWce3awdeo1*so lm2*
UTC01_03	Caeser cipher	Cee ihrasrcpe*

Case Study:

Core Java Problem Statement using JDBC

Problem Description/Current System

Design and implement a library management application for a pre-university. It will be used by the librarian at the university, to manage the transaction starting from adding a book to the database to the return of a book

System Requirement Specification

The main menu contains the following options. The details for each option are given below.

1. Add a Book
2. Borrow a Book
3. Return a Book
4. Exit

Add a Book

The application allows the librarian to add book details. The librarian is prompted to provide the details of a book here. The sample responses are given in red colour.

The following rules should be incorporated in the implementation:

- ISBN should be a 10-digit number.
- Publication Date should be earlier then current date.
- The status of the book is set to "Available" by default

```
Main Menu
    1. Add a Book
    2. Borrow a Book
    3. Return a Book
    4. Exit

Please choose an option [1, 2, 3, 4]:  1

Enter the title of the book: C# Programming

Enter the publication date(DD-MM-YYYY):15-06-2004

Enter the author name: Steve Holzner

Enter the ISBN: 3546712392

Enter the category of the book("Technical", "management", "General"): Technical

Book added successfully
```

The system saves the details entered by the librarian with status of book as 'Available' and displays the message "Book added successfully".

Borrow a book

The application allows the librarian to issue a book. The librarian should choose the search criteria and then choose the listed books availalabe. The sample responses are given in red color

The following rules should be incorporated in the implementation:

- Books with only status 'Available' should be listed for reservation.
- On successful reservation, the status of the book should be changed to 'Issued'

```
Main Menu:
    1. Add a Book
    2. Borrow a Book
    3. Return a Book
    4. Exit
Please choose an option [1, 2, 3, 4]: 2
```

```
Search criteria:
    1. Search by Title
    2. Search by Author
    3. Search by category
Please choose a search criteria [1, 2, 3, 4]: 1
Enter the title of the book : c#
There are 3 books matching your search criteria:
    1. C# programming
    2. C# 3.0
    3. Inside C#

Which of the listed books do you wish to borrow? [1, 2, 3, 4]: 2
The book 'C# 3.0' issued successfully.
```

```
Do you wish to make one more reservation? [ Y or N]=Y
Search criteria:
    1. Search by Title
    2. Search by Author
    3. Search by category
Please choose a search criteria [1, 2, 3, 4]: 2
Enter the author name : Chetan bhagat
There are 3 books matching your search criteria:
    1. Five point someone
    2. 2 states
    3. One night at call center
Which of the listed books do you wish to borrow? [1, 2, 3, 4]: 3
The book 'One night at call center' issued successfully ...

Do you wish to make one more reservation? [ Y or N]=Y
Search criteria:
    1. Search by Title
    2. Search by Author
    3. Search by category
Please choose a search criteria [1, 2, 3, 4]: 3
Enter category : Technical
There are 2 books under category 'Technical':
    1. java
    2. ADO.net
Which of the listed books do you wish to borrow? [1, 2, 3, 4]: 1
The book 'java' issued successfully...

Do you wish to make one more reservation? [ Y or N]=N
```

The system updates the status of the book to 'Issued' and displays the message "The book <book name> issued successfully".

Return a Book

The issued book can be returned using 'Return a Book' usecase. The librarian should enter the International Standard Book Number (ISBN) to return a book.
The following rules should be incorporated in the implementation:

- On successful return, the book status should be changed to 'Available'

```
Main Menu:
    1. Add a Book
    2. Borrow a Book
    3. Return a book
    4. Exit
Please choose an option [1, 2, 3, 4]: 4

Enter the ISBN: 3546712392

The book returned successfully.
```

Exit

The user can choose to exit the application. The sample responses are given in red color

```
Main Menu:
    1. Add a Book
    2. Borrow a Book
    3. Return a Book
    4. Exit
Please choose an option [1, 2, 3, 4]: 6
        The option chosen by you is invalid. Please retry.
Main Menu:
    1. Add a Book
    2. Borrow a Book
    3. Return a Book
    4. Exit
Please choose an option [1, 2, 3, 4]: 4
        Application closed.
```